

A finite strain visco-hyperelastic damage model for rubber-like materials: theory and numerical implementation

Zhenjiang Du¹, Yan Yang¹, Zhongtong Wang², Xinggui Fan³, and Tongqing Lu^{1*}

¹ State Key Lab for Strength and Vibration of Mechanical Structures, Department of Engineering Mechanics, Xi'an Jiaotong University, Xi'an 710049, China;

² Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca 14853, United States;

³ Navigation and Control Technology Research Institute of China North Industries Group Corporation, Beijing 100089, China

Received November 27, 2022; accepted January 13, 2023; published online March 8, 2023

Many rubber-like materials exhibit hyperelastic, time-dependent, rate-dependent and progressive damage behaviors. In our previous work (Lu et al., 2020), we proposed a hyperelastic damage model to characterize the strain-softening behavior of soft materials. The model modifies the strain energy function of a single chain by introducing an internal damage variable D and then maps the deformation of chains to the macroscopic deformation. In this work, we extend this model to incorporate the time-dependent viscous effect using the Prony series-based nonlinear theory. We further implement the finite strain visco-hyperelastic damage model into finite element software ABAQUS by a user material subroutine UMAT. We use the experimental data of a kind of acrylic polymer under uniaxial tension in literature to calibrate the model parameters, including 4 time-independent parameters and 6 time-dependent parameters. We then use the calibrated parameters to simulate the uniaxial tension and stress relaxation of the acrylic polymer specimen with a complex geometry. The simulated results agree with the experimental data with a remarkable accuracy.

Soft materials, Hyperelastic, Viscoelastic, Damage, UMAT

Citation: Z. Du, Y. Yang, Z. Wang, X. Fan, and T. Lu, A finite strain visco-hyperelastic damage model for rubber-like materials: theory and numerical implementation, *Acta Mech. Sin.* **39**, 222473 (2023), <https://doi.org/10.1007/s10409-023-22473-x>

1. Introduction

Rubber-like soft materials are widely used in numerous applications including soft robots [1,2], stretchable electronics [3], and bridge bearings [4-6]. Subject to loads, irreversible damage can accumulate in many soft materials. Meanwhile, soft materials usually exhibit time-dependent viscous behavior. It is of great interest for researchers to simultaneously consider viscoelasticity and damage of soft materials.

In modelling damage, a commonly used phenomenological approach is to use a damage variable with a certain physical meaning, such as modules or volume fraction of void, to capture the macroscopic damage behavior based on

continuum damage mechanics [7-12]. The evolution of the damage variable is obtained by extensive experimental calibration. To account for the finite deformation of soft materials, the microscopic damage model on the basis of the evolution of polymer chains has been developed [13]. The model prescribes the evolution of the average quantity of monomer segments in the chain and the chain density in the loading process. Following this approach, many constitutive models have been proposed for different materials with different underlying damage mechanisms, such as interpenetrating network elastomers [14], nanocomposites [15], double-network hydrogels [16], multiple network elastomers [17], and self-healing gels [18]. For the ease of implementation in finite element (FE) methods, the most widely used model is the pseudo elasticity model which constructs a single damage variable to evolve with deformation [19]. This model has been adopted by commercial FE software [20].

*Corresponding author. E-mail address: tongqinglu@mail.xjtu.edu.cn (Tongqing Lu)
Executive Editor: Yuli Chen

In modelling viscoelasticity, the most widely used model to describe the linear-viscoelastic behavior is Maxwell model, Kelvin-Voigt model, and the Boltzmann superposition principle [21-23]. To account for the finite deformation of soft materials, the hereditary integral approach is developed to describe the viscous effect [24-26]. Another commonly used approach is to split the free energy into elastic and inelastic contributions. This framework was initially proposed by Green and Tobolsky [27] and has been largely explored to describe viscoelasticity. Following this framework, some researchers developed phenomenological models [28-30] and some researchers developed micro-mechanics motivated models [31-35].

In modelling damage and viscoelasticity simultaneously, researchers simply combine the damage model and viscoelasticity model. For example, Özüpek and Becker [36] developed a constitutive model taking into account damage and nonlinear viscoelastic behaviors. The microstructural damage was described by internal variables evolution (void volume) and the viscous effect was described by the Prony series method [26]. They further generalized this model to hyperelastic viscoelastic damage model, and then implemented the model in FE software [37,38]. Sasso et al. [39] used the modified Ogden-Roxburgh formulation to describe Mullins effect and the Prony series to capture the stress relaxation of rubber. Lu et al. [40] introduced a damageable spring to the Zenor model to model the rate-dependent Mullin's effect of tough hydrogels.

In our previous work, we developed a hyperelastic damage model to capture the strain-softening behavior of soft materials [41]. The model modifies the strain energy function of a single chain by introducing an internal damage variable and then maps the deformation of chains to the macroscopic deformation using eight chain model. The model has two distinct advantages. First, the model has only one additional parameter to characterize damage which is easy to calibrate with experiment. Second, the formulation of the model, similar to the pseudo elasticity form of Ogden's model [19], is very convenient to implement into FE methods. In this work, we extend our model [41] to incorporate the viscous effect using the Prony series-based nonlinear theory [26]. We implement the model into ABAQUS by a user material subroutine UMAT and find that the simulated results agree with the experimental data very well.

2. The finite strain visco-hyperelastic damage model

The finite deformation model incorporating viscoelasticity and damage is formulated as follows. Define the deformation gradient of a representative volumetric element as $\mathbf{F} = \partial \mathbf{x} / \partial \mathbf{X}$. Define the distortion gradient as $\bar{\mathbf{F}} = J^{-1/3} \mathbf{F}$

where $J = \det(\mathbf{F})$. The left Cauchy-Green deformation tensor is $\mathbf{B} = \mathbf{F}\mathbf{F}^T$. The distortion gradient of the Cauchy-Green deformation tensor is $\bar{\mathbf{B}} = J^{-2/3} \mathbf{B}$.

The total Kirchhoff stress is divided into the deviatoric part and the hydrostatic part:

$$\boldsymbol{\tau}(t) = \boldsymbol{\tau}^D(t) + \boldsymbol{\tau}^H(t). \quad (1)$$

Following Refs. [20] and [26], the deviatoric part and the hydrostatic part are calculated as

$$\boldsymbol{\tau}^D(t) = \boldsymbol{\tau}_0^D(t) + \text{dev} \left\{ \int_0^\xi \frac{\partial g_R(\xi')}{\partial \xi'} \left[\mathbf{F}_t^{-1}(t-t') \boldsymbol{\tau}_0^D(t-t') \mathbf{F}_t^{-T}(t-t') \right] d\xi' \right\}, \quad (2)$$

$$\boldsymbol{\tau}^H(t) = \boldsymbol{\tau}_0^H(t), \quad (3)$$

where $\boldsymbol{\tau}_0^D$ is the instantaneous part of $\boldsymbol{\tau}^D(t)$, $\boldsymbol{\tau}_0^H$ is the instantaneous part of $\boldsymbol{\tau}^H(t)$, $g(\xi')$ is the shear relaxation modules, $\text{dev}(\cdot)$ is the deviatoric operators, $(\cdot)^{-1}$ and $(\cdot)^{-T}$ indicate the inverse and the transposer inverse tensor, and $\bar{\mathbf{F}}_t(t-t') = \frac{\partial \mathbf{x}(t-t')}{\partial \mathbf{x}(t)}$.

The relaxation modulus in Eq. (2) is represented by Prony series as

$$g_R(t) = g_\infty + \sum_{i=1}^N g_i \exp\left(-\frac{t}{\tau_i}\right), \quad (4)$$

where g_∞ is the infinite value of modulus, g_i is Prony coefficient, τ_i is relaxation time.

The instantaneous part of the stress $\boldsymbol{\tau}_0^D$ and $\boldsymbol{\tau}_0^H$ are obtained from the hyperelastic free energy function. Here we use the pseudo-elasticity model proposed in our previous work [41]. This model introduces an internal variable to characterize damage and combines with the eight-chain network model. Specifically, the free energy density function is

$$W = NkTn \left[\frac{\lambda}{\sqrt{n}D} \beta + \ln \left(\frac{\beta}{\sinh(\beta)} \right) + \Phi(D) \right] + Y(J-1)^2, \quad (5)$$

where N is the chain density, k is the Boltzmann constant, T is the temperature, n is the number of links in a chain, λ is the stretch of each chain, $\beta = \ell^{-1} \left(\frac{\lambda}{n} \right)$, $\ell^{-1}(\cdot)$ is the inverse Langevin function, and Y is the bulk modulus. The eight-chain model relates the stretch of each chain with the macroscopic deformation by $\lambda = \sqrt{\text{tr}(\bar{\mathbf{B}})/3}$. In this paper, we use the function $\Phi(D) = \kappa(D-1)$ to represent the damage behavior of the material. The additional damage parameter κ controls the softening of the stress-stretch curve. According to the function $\frac{\partial W(\mathbf{F}, D)}{\partial D} = 0$, the internal

variable evolves with deformation as

$$D = \sqrt{\frac{\sqrt{n}A}{\kappa}}\beta. \quad (6)$$

Then the instantaneous Kirchhoff stress can be obtained as

$$\tau_0 = \frac{NkT\sqrt{n}}{3} \frac{\beta}{AD} \bar{\mathbf{B}} + 2JY(J-1)\mathbf{I}. \quad (7)$$

where $\mathbf{I}$ is the unit tensor.

3. Numerical implementation

The constitutive model was implemented in ABAQUS via user subroutine UMAT. We deduce the stress tensor update and the tangent stiffness update [20].

3.1 Discrete stress-strain relationship

The deviatoric part of the stress is

$$\tau^D(t) = \tau_0^D(t) - \text{dev} \left[\sum_{i=1}^N \frac{g_i}{\tau_i} \cdot \int_0^\xi \bar{\mathbf{F}}_t^{-1}(t-t') \tau_0^D(t-t') \bar{\mathbf{F}}_t^{-T}(t-t') \exp \left(-\frac{\xi'}{\tau_i} \right) d\xi' \right]. \quad (8)$$

In Eq. (8), the internal stress τ_i^D , related to each term of the series is defined as follows [20]:

$$\tau_i^D(t) = \frac{g_i}{\tau_i} \int_0^\xi \bar{\mathbf{F}}_t^{-1}(t-t') \tau_0^D(t-t') \bar{\mathbf{F}}_t^{-T}(t-t') \exp \left(-\frac{\xi'}{\tau_i} \right) d\xi'. \quad (9)$$

It is assumed that all the quantities (e.g., stress and strain) at a time t are already acquired, and after mathematical manipulations, all internal stress quantities are known at the time $t + \Delta t$ as

$$\tau_i^D(t + \Delta t) = \alpha_i g_i \tau_0^D(t + \Delta t) + \beta_i g_i \hat{\tau}_0^D(t) + \gamma_i \hat{\tau}_i^D(t), \quad (10)$$

where

$$\gamma_i = \exp \left(-\frac{\Delta t}{\tau_i} \right), \quad \alpha_i = 1 - \frac{\tau_i}{\Delta t} (1 - g_i), \quad \beta_i = \frac{\tau_i}{\Delta t} (1 - g_i) - g_i. \quad \text{The}$$

stress $\hat{\tau}_i^D(t)$ and $\hat{\tau}_0^D(t)$ are

$$\hat{\tau}_i^D(t) = \bar{\mathbf{F}}_t(t + \Delta t) \tau_i^D(t) \bar{\mathbf{F}}_t^T(t + \Delta t), \quad i = 0, 1, \dots, s, \quad (11)$$

$$\hat{\tau}_0^D(t) = \bar{\mathbf{F}}_t(t + \Delta t) \tau_0^D(t) \bar{\mathbf{F}}_t^T(t + \Delta t), \quad (12)$$

where s is the quantity of the Prony series coefficient.

The hydrostatic part of the stress $\tau_0^H(t + \Delta t)$ is updated the same as $\tau_0^D(t)$. Finally, using the instantaneous viscoelastic theory, the discrete form of the total Kirchhoff stress reads

$$\tau(t + \Delta t) = \tau_0^D(t + \Delta t) - \sum_{i=1}^N \tau_i^D(t + \Delta t) + \tau_0^H(t + \Delta t). \quad (13)$$

3.2 Tangent stiffness

We begin with Oldroyd stress rate of our model's elastic

component [42] which defined as

$$\begin{aligned} \wp_v(\tau_0) &= \dot{\tau}_0 - \mathbf{L} \cdot \tau_0 - \tau_0 \cdot \mathbf{L}^T \\ &= 4\mathbf{B} \frac{\partial^2 W(\mathbf{B})}{\partial \mathbf{B}^2} \mathbf{B} : \mathbf{D} := \mathbf{C}^{\text{EQ}} : \mathbf{D}, \end{aligned} \quad (14)$$

where $\mathbf{L}$ is the velocity gradient tensor, $\mathbf{D} = \frac{1}{2}(\mathbf{L} + \mathbf{L}^T)$ is the deformation rate tensor, $\mathbf{C}^{\text{EQ}}$ is the elastic tangent modulus and can be decomposed into the deviatoric part and hydrostatic parts.

$$\mathbf{C}^{\text{EQ}} = \mathbf{C}_{\text{vol}}^{\text{EQ}}(J) + \mathbf{C}_{\text{iso}}^{\text{EQ}}(\bar{\mathbf{B}}). \quad (15)$$

The deviatoric part for elastic tangent modulus reads

$$\begin{aligned} \mathbf{C}_{\text{iso}}^{\text{EQ}}(\bar{\mathbf{B}}) &= 4\mathbf{B} \frac{\partial^2 W^{\text{iso}}(\bar{\mathbf{B}})}{\partial \bar{\mathbf{B}}^2} \mathbf{B} \\ &= \frac{2}{3} \text{tr}(\bar{\boldsymbol{\tau}}) \mathbf{P} - \frac{2}{3} (\tau_0^D \otimes \mathbf{I} + \mathbf{I} \otimes \tau_0^D) + \mathbf{P} : \bar{\boldsymbol{\tau}} : \mathbf{P}, \end{aligned} \quad (16)$$

where $\bar{\boldsymbol{\tau}}$ is an algorithmic tangent modulus and has the following expression:

$$\bar{\boldsymbol{\tau}} = 4\mathbf{B} \frac{\partial^2 W^{\text{iso}}(\bar{\mathbf{B}})}{\partial \bar{\mathbf{B}}^2} \bar{\mathbf{B}}, \quad (17)$$

and the hydrostatic part for elastic tangent modulus is

$$\begin{aligned} \mathbf{C}_{\text{vol}}^{\text{EQ}}(J) &= 4\mathbf{B} \frac{\partial^2 W^{\text{vol}}(J)}{\partial \mathbf{B}^2} \mathbf{B} \\ &= \left(\frac{\partial^2 W^{\text{vol}}(J)}{\partial J^2} + J \frac{\partial W^{\text{vol}}(J)}{\partial J} \right) \mathbf{I} \otimes \mathbf{I} - 2J \frac{\partial W^{\text{vol}}(J)}{\partial J} \mathbf{I}. \end{aligned} \quad (18)$$

In this work, the algorithmic tangent modulus is

$$\bar{\boldsymbol{\tau}} = 4\mathbf{B} \frac{\partial^2 W^{\text{iso}}(\bar{\mathbf{B}})}{\partial \bar{\mathbf{B}}^2} \bar{\mathbf{B}} = -2\mathbf{B} \left(\frac{NkT\sqrt{n}}{AD} \beta \right) \bar{\mathbf{B}}. \quad (19)$$

According to Ref. [20], the viscoelastic tangent modulus $\mathbf{C}^{\text{NON}}$ is also divided into the viscoelastic deviatoric moduli and viscoelastic hydrostatic moduli through

$$\mathbf{C}^{\text{NON}} = \mathbf{C}_{\text{vol}}^{\text{NON}}(J) + \mathbf{C}_{\text{iso}}^{\text{NON}}(\bar{\mathbf{B}}). \quad (20)$$

Following Ref. [20], the viscoelastic hydrostatic moduli is

$$\mathbf{C}_{\text{vol}}^{\text{NON}}(J) = \mathbf{C}_{\text{vol}}^{\text{EQ}}(J), \quad (21)$$

and the viscoelastic deviatoric part is

$$\mathbf{C}_{\text{iso}}^{\text{NON}}(\bar{\mathbf{B}}) = \left(1 - \sum_{i=1}^r \alpha_i g_i \right) \mathbf{C}_{\text{iso}}^{\text{EQ}}(\bar{\mathbf{B}}). \quad (22)$$

In order to obtain the Jaumann derivative used in ABAQUS, the following transformation was used [43]:

$$\mathbf{C}_{abcd}^J = \mathbf{C}_{abcd}^{\text{NON}} + \frac{1}{2} (\delta_{ac} \tau_{bd} + \tau_{ac} \delta_{bd} + \delta_{ad} \tau_{bc} + \tau_{ad} \delta_{bc}), \quad (23)$$

where δ is Kronecker symbol and the superscript $(\cdot)^J$ refers to Jaumann. At last, the tangent modulus implemented in the UMAT is $\mathbf{C}^{\text{ABAQUS}} = \frac{1}{J} \mathbf{C}^J$. With the above stress update and tangent stiffness derivation results, we will be able to write UMAT. The accuracy verification of the UMAT subroutine is given in [Appendix](#).

4. Comparison with experimental results

4.1 Parameter calibration and discussion

We use the experimental data of VHB (a commercial brand of acrylic polymer, short for “very high bond”) under uniaxial tension and stress relaxation extracted from Ref. [44]. We fit the stress relaxation data to obtain the time dependent parameters in Eq. (4) using non-linear least squares method. The fitting results are shown in Fig. 1a. The fitting error is within 2%. We then fit the uniaxial tension data along with the obtained time dependent parameters to obtain the time independent parameters in Eq. (7). The fitting results are shown in Fig. 1b. We have also compared different numbers of time-dependent parameters for the Prony series g_i , τ_i . With $N = 2$, we have in total 4 time-dependent parameters and the corresponding fitting error in stress-time curves is greater than 10%. With $N = 3$, we have in total 6 time-dependent parameters and the corresponding fitting error in stress-time curves within 2% (Fig. 1b), which satisfies most applications. With a total of ten parameters, the fitting accuracy between theory and experiment under different stretch rates of 0.1/s, 0.05/s and 0.01/s is remarkable. The calibrated parameters are listed in Table 1.

4.2 Uniaxial tension and stress relaxation of VHB specimen with a complex geometry

We use the experimental data of VHB specimens with a complex geometry extracted from Refs. [44,45]. We implemented the model in Abaqus/Standard (2020) by writing UMAT. The configuration and the mesh of the specimen in the undeformed state are shown in Fig. 2a which was discretized with 2241 C3D8 elements. The thickness was only

with a single element. We first consider the case of uniaxial tension. In the FE simulation, the boundary conditions are set as follows. The bottom face is pinned and the uniaxial displacement is imposed on the top face along 2 direction. The cumulative reaction force is obtained by summarizing the reaction force on the bottom face in the 2 direction. All leftover faces have no traction. Alkhoury et al. [44] presented two snapshots of uniaxial tension measured by digital image correlation (DIC) when the applied displacement is 12 mm and 19.4 mm respectively (Fig. 2b and c). We also plot the simulated results of logarithm strain LE_{22} using the UMAT subroutine and the parameters in Table 1 (Fig. 2b and c). The agreement between the DIC results and the simulation is good. Figure 2d and e shows the contour plot of damage for the VHB specimen at the prescribed displacements 12 mm and 19.4 mm. We next consider the case of stress relaxation. Alkhoury et al. [44] applied a constant displacement to the VHB specimen and then held it for 300 s (Fig. 3b). The measured force as a function of displacement and time are plotted as dots in Fig. 3c and d respectively. In our FE simulation, we apply the same loading history and calculate the force as a function of displacement and time as the solid line in Fig. 3c and d respectively. The agreement between the experimental data and the simulation is remarkable.

5. Conclusion

We develop a finite strain visco-hyperelastic damage model by extending our previous hyperelastic damage model to incorporate viscoelasticity. The viscoelasticity is modelled by the Prony series-based viscoelastic theory. We further

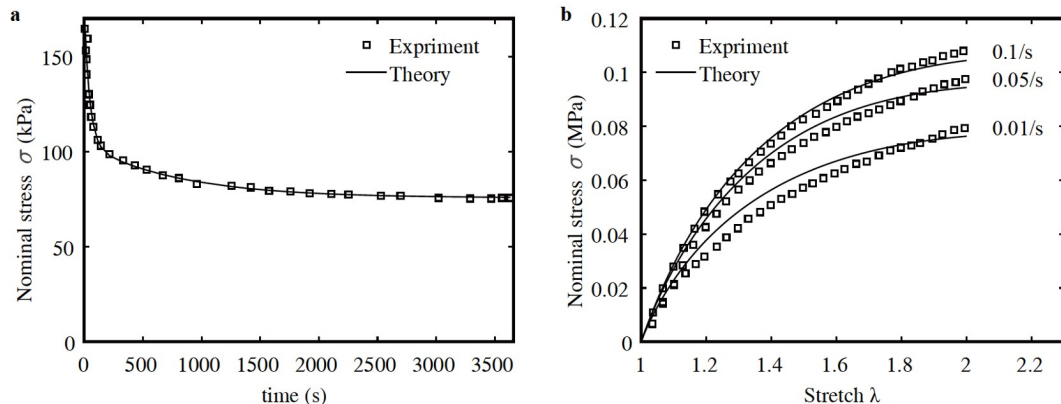


Figure 1 Parameter calibration using the experimental data of VHB specimen. **a** Time-dependent parameters are calibrated from the experimental data of uniaxial stress relaxation. **b** Time-independent parameters are calibrated from the experimental data of uniaxial tension tests under three different loading rates of 0.01/s, 0.05/s, and 0.1/s.

Table 1 Calibrated parameters for VHB (experimental data extracted from Ref. [44])

Time-independent parameters				Time-dependent parameters					
NkT (MPa)	n	Y (MPa)	κ	g_1	g_2	g_3	τ_1 (s)	τ_2 (s)	τ_3 (s)
0.115	200	1×10^5	3	0.201	0.213	0.184	1.12	10.15	101.36

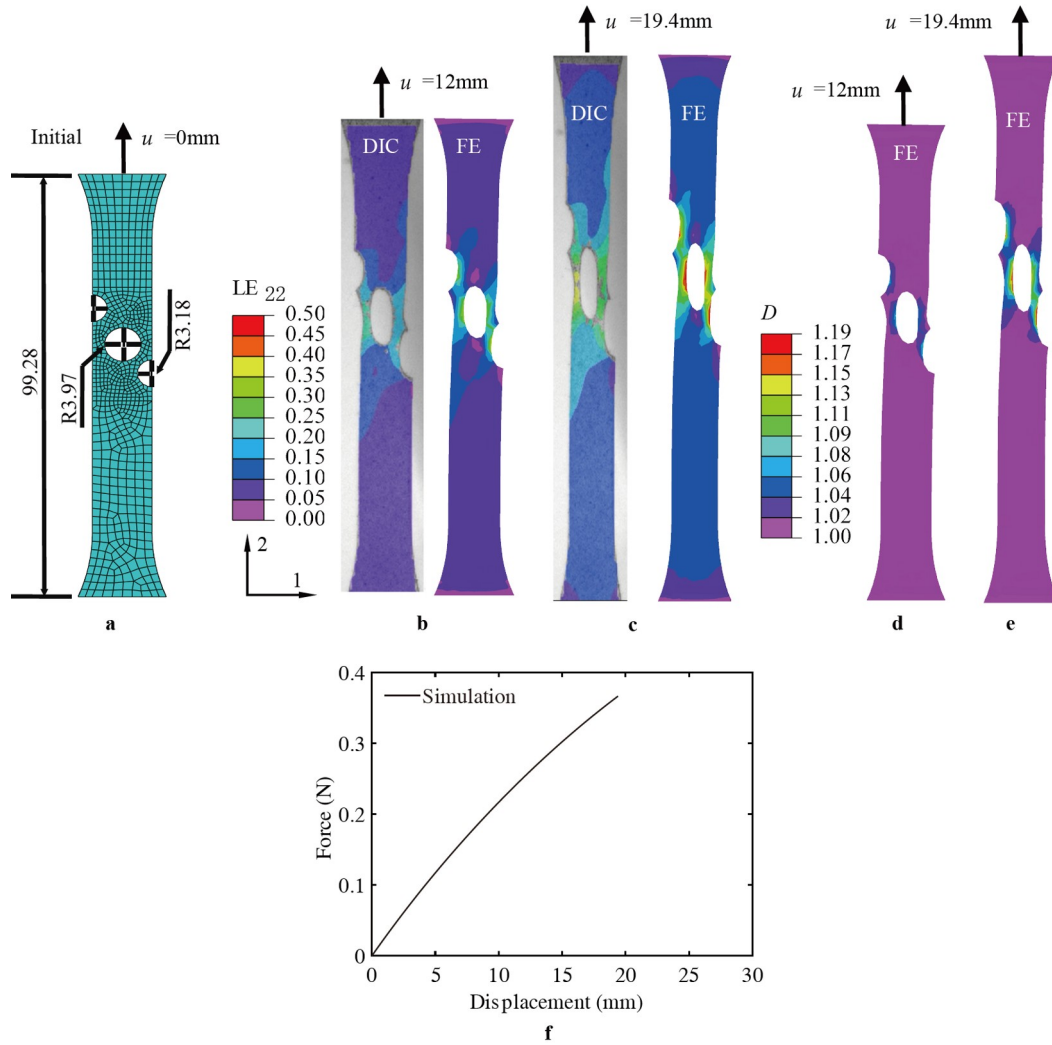


Figure 2 Comparison between the strain field experimentally measured by DIC [44] and the FE results for a VHB specimen with complex geometry. **a** Configuration in the undeformed state. The comparison is shown at prescribed displacements of **b** 12 mm and **c** 19.4 mm. The damage distribution of the specimen under the displacement of **d** 12 mm and **e** 19.4 mm. **f** Calculated force-displacement curve.

implement the model into FE software ABAQUS via UMAT. After a parameter calibration with the experimental data of uniaxial tension of VHB, the model predicts the inhomogeneous deformation of a VHB specimen with a complex geometry under tension and stress relaxation. The simulated results agree with the experimental data with remarkable accuracy. The present model has a simple form and the number of parameters are few, which is useful for FE simulations on complicated engineering applications.

Appendix: Numerical implementation (UMAT) validation

The discrete stress-strain relationship and the tangent stiffness formulation are summarized in Sect. 3.1 and 3.2. The three-dimensional hyperelastic, finite strain viscoelastic, damage model is implemented into ABAQUS by develop-

ing a user-defined material subroutine (UMAT). To test the UMAT, we calculate two cases of homogeneous deformation by applying only one element (C3D8 element), uniaxial tension and simple shear and compare the FE results with the theoretical results. The convergence of the visco-hyperelastic model in the FE implementation is verified by changing the increment size.

A1 Homogeneous uniaxial tension

For uniaxial tension, the nonzero stress component is σ . The principal stretch in the loading direction is $\lambda_1(t) = \lambda(t)$. Consider a loading history defined by

$$\lambda(t) = 1 + \dot{\lambda}t, \quad (\text{A1})$$

where $\dot{\lambda}$ represents the stretch rate. The material is assumed to be incompressible, giving $\lambda_2 = \lambda_3 = \lambda^{-1/2}$. Then the instantaneous Cauchy stress $\sigma(t)$ at $t = 0$ can be calculated as

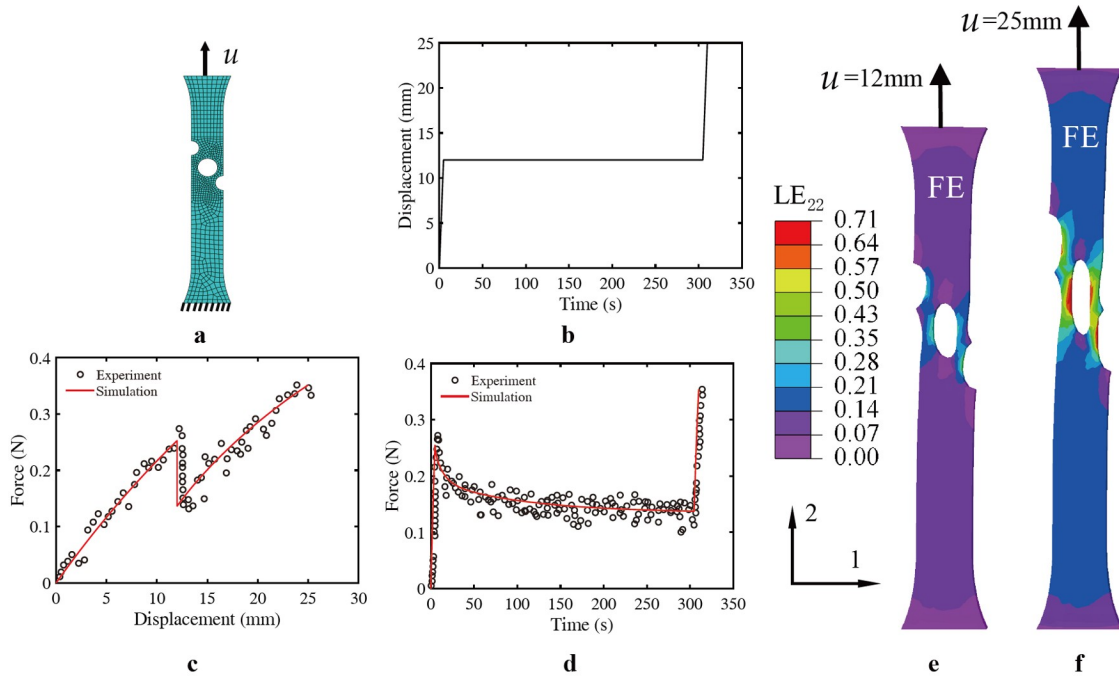


Figure 3 Comparison between the experimentally measured force-displacement response [44] and the FE results for a VHB specimen with complex geometry. **a** Configuration in the undeformed state. **b** Loading history (under uniaxial tension). **c** Comparison of force-displacement curve. **d** Comparison of force-time curve. The calculated strain field of the specimen under the displacement of **e** 12 mm and **f** 25 mm.

$$\sigma_0(t) = \frac{NkT\sqrt{n}}{3J} \frac{\beta}{AD} (\lambda^2 - \lambda^{-1}). \quad (A2)$$

Following Eqs. (1)-(3), the total Cauchy stress can be expressed as

$$\sigma(t) = \sigma_0(t) - \sum_{i=1}^3 \frac{g_i}{\tau_i} \int_0^t \frac{\lambda^2(t-t')}{\lambda^2(t)} \sigma_0(t-t') \exp\left(-\frac{\zeta'}{\tau_i}\right) d\zeta'. \quad (A3)$$

Following Eq. (13), the updated stress is obtained as

$$\sigma(t_{n+1}) = \sigma_0(t_{n+1}) - \sum_{i=1}^3 \sigma^i(t_{n+1}), \quad (A4)$$

$$\sigma^i(t_{n+1}) = \frac{\lambda^2(t_{n+1})}{\lambda^2(t_n)} (\beta_i g_i \tau_0(t_n) + \gamma_i \sigma^i(t_n)) + \alpha_i g_i \sigma_0(t_{n+1}), \quad (A5)$$

where σ^i is the viscoelastic constituent of the deviatoric part of the Cauchy stress.

Combining Eqs. (A1)-(A5), we plot the stress-stretch curves under three stretch rates 1/s, 0.1/s, and 0.01/s in Fig. A1a. In the calculation and the subsequent case studies (Sect. A2), we use the material parameter as follows, $NkT = 0.1$ MPa, $n = 225$, $Y = 1 \times 10^5$ MPa, $\kappa = 3$, $g_1 = 0.09$, $g_2 = 0.08$, $g_3 = 0.07$, $\tau_1 = 1$ s, $\tau_2 = 10$ s, $\tau_3 = 100$ s. Figure A1b shows the stress relaxation under uniaxial tension, that is, the material is suddenly stretched to $\lambda = 2, 3, 4$ and then holds 1000 s to calculate the stress evolution. As expected, the FE results perfectly match the MATLAB results.

A2 Homogeneous simple shear

For simple shear, the nonzero stress component is

$\sigma_{12}(t) = \tau(t)$. The shear strain in the loading direction is $\gamma_{12}(t) = \gamma(t)$. We consider a ramp stretch history defined by $\gamma(t) = \dot{\gamma}t$, (A6)

where $\dot{\gamma}$ is the shear rate. Following Eqs. (1)-(3), the total shear stress can be expressed as

$$\tau(t) = \tau_0(t) - \sum_{i=1}^3 \frac{g_i}{\tau_i} \int_0^t \tau_0(t-t') \exp\left(-\frac{\zeta'}{\tau_i}\right) d\zeta'. \quad (A7)$$

Following Eq. (13), the discrete form of shear stress reads

$$\tau(t_{n+1}) = \tau_0(t_{n+1}) - \sum_{i=1}^3 \tau^i(t_{n+1}), \quad (A8)$$

$$\tau^i(t_{n+1}) = (\beta_i g_i \tau_0(t_n) + \gamma_i \tau^i(t_n)) + \alpha_i g_i \tau_0(t_{n+1}), \quad (A9)$$

where τ^i is the viscoelastic constituent of the deviatoric part of the Cauchy stress.

Combining Eqs. (A6)-(A9), we plot the stress-stretch curves under three stretch rates 1/s, 0.1/s, and 0.01/s in Fig. A1c. Figure A1d shows the stress relaxation under simple shear, that is, the material is suddenly sheared to the strain $\gamma = 0.25, 0.5, 1$ and then hold 1000 s to calculate the stress evolution. The FE results perfectly match the MATLAB results.

A3 Convergence verification

We vary the increment size 0.008, 0.01 and 0.08 to test the convergence. The results show that the stress-strain curves

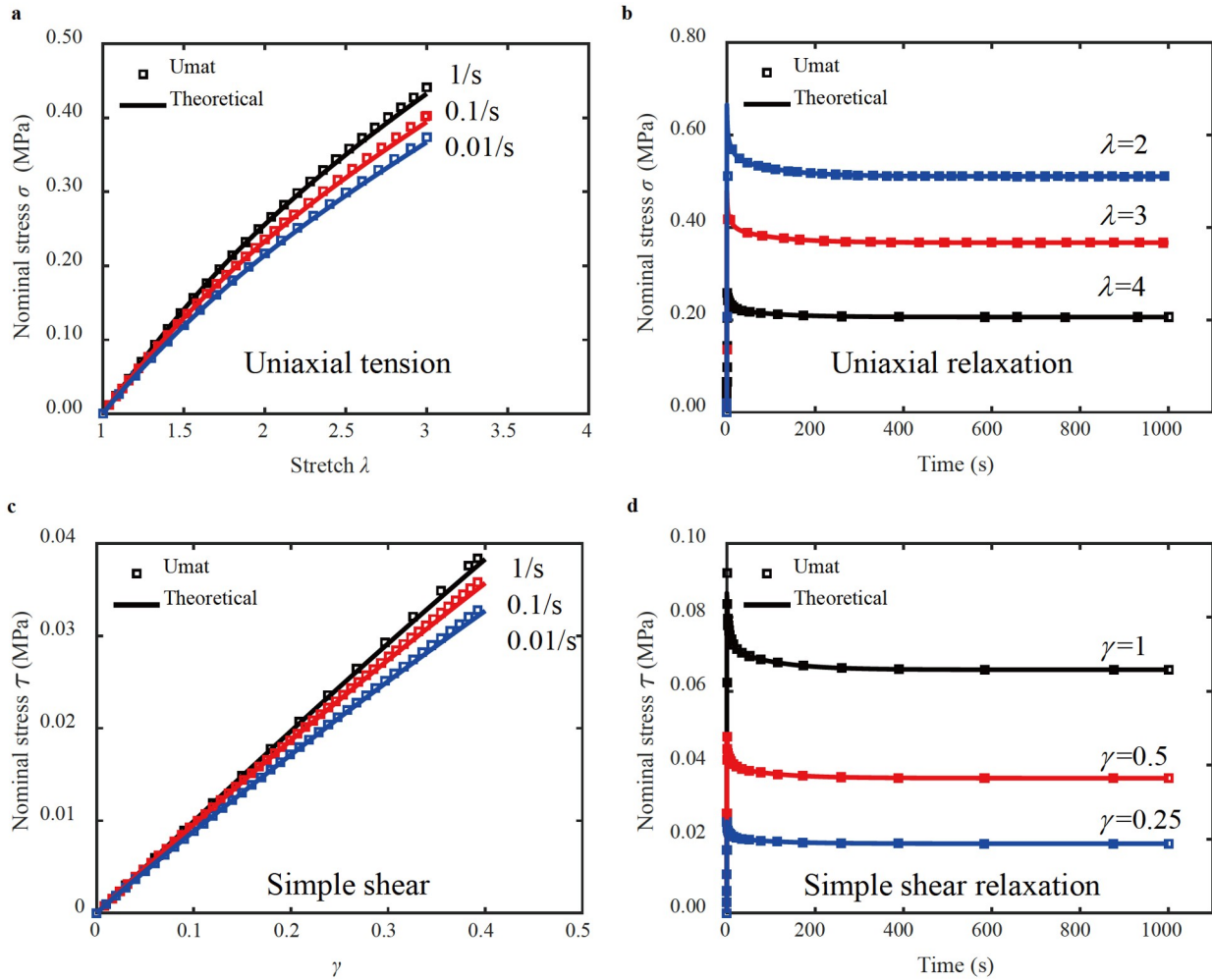


Figure A1 Comparison between theoretical results and FE results in modelling homogeneous deformation. **a** Homogeneous uniaxial tension. **b** Homogeneous uniaxial relaxation. **c** Homogeneous simple shear. **d** Homogeneous simple shear relaxation.

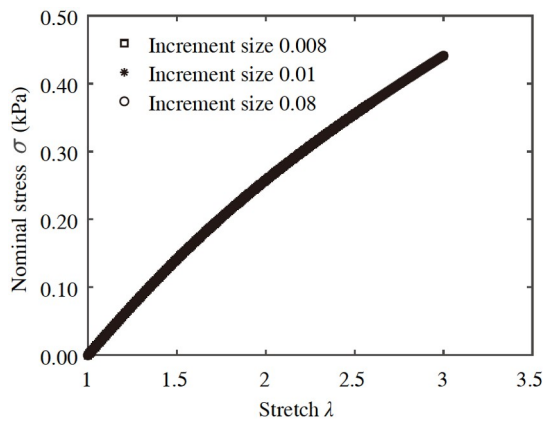


Figure A2 Stress-stretch curves of uniaxial tension using different increment sizes.

of uniaxial tension overlap completely with different increment sizes (Fig. A2).

Author contributions Zhenjiang Du created the constitutive model, achieved the UMAT, processed the numerical simulations and wrote the

first draft of the manuscript. Yang Yan, Zhongtong Wang and Xinggui Fan revised the final version. Tongqing Lu supervised the whole project and revised the manuscript.

Acknowledgements This work was supported by the National Natural Science Foundation of China (Grant No. 11922210).

- 1 P. Calvert, Hydrogels for soft machines, *Adv. Mater.* **21**, 743 (2009).
- 2 S. Coyle, C. Majidi, P. LeDuc, and K. J. Hsia, Bio-inspired soft robotics: Material selection, actuation, and design, *Extreme Mech. Lett.* **22**, 51 (2018).
- 3 C. Yang, and Z. Suo, Hydrogel ionotronics, *Nat. Rev. Mater.* **3**, 125 (2018).
- 4 C. S. Tsai, T. C. Chiang, B. J. Chen, and S. B. Lin, An advanced analytical model for high damping rubber bearings, *Earthquake Engng. Struct. Dyn.* **32**, 1373 (2003).
- 5 A. R. Bhuiyan, Y. Okui, H. Mitamura, and T. Imai, A rheology model of high damping rubber bearings for seismic analysis: Identification of nonlinear viscosity, *Int. J. Solids Struct.* **46**, 1778 (2009).
- 6 E. Tubaldi, S. A. Mitoulis, and H. Ahmadi, Comparison of different models for high damping rubber bearings in seismically isolated bridges, *Soil Dyn. Earthquake Eng.* **104**, 329 (2018).
- 7 L. M. Kachanov, Time of the rupture process under creep conditions, *Izy Akad. Nank SSR Otd Tech Nauk* **8**, 26 (1958).
- 8 J. Lemaitre, A continuous damage mechanics model for ductile

- fracture, Trans. Asme J. Eng. Mater. Technol. **107**, 83 (1985).
- 9 T. J. Wang, Unified CDM model and local criterion for ductile fracture—I. Unified CDM model for ductile fracture, *Eng. Fract. Mech.* **42**, 177 (1992).
 - 10 T. J. Wang, Unified CDM model and local criterion for ductile fracture—II. Ductile fracture local criterion based on the CDM model, *Eng. Fract. Mech.* **42**, 185 (1992).
 - 11 F. Xu, N. Aravas, and P. Sofronis, Constitutive modeling of solid propellant materials with evolving microstructural damage, *J. Mech. Phys. Solids* **56**, 2050 (2008).
 - 12 M. Lei, J. Wang, J. Cheng, J. Xiao, L. Wen, H. Lu, and X. Hou, A constitutive model of the solid propellants considering the interface strength and dewetting, *Compos. Sci. Tech.* **185**, 107893 (2020).
 - 13 G. Marckmann, E. Verron, L. Gornet, G. Chagnon, P. Charrier, and P. Fort, A theory of network alteration for the Mullins effect, *J. Mech. Phys. Solids* **50**, 2011 (2002).
 - 14 X. Zhao, A theory for large deformation and damage of interpenetrating polymer networks, *J. Mech. Phys. Solids* **60**, 319 (2012).
 - 15 Q. Wang, and Z. Gao, A constitutive model of nanocomposite hydrogels with nanoparticle crosslinkers, *J. Mech. Phys. Solids* **94**, 127 (2016).
 - 16 P. Zhu, and Z. Zhong, Modelling the mechanical behaviors of double-network hydrogels, *Int. J. Solids Struct.* **193-194**, 492 (2020).
 - 17 R. Xiao, N. Han, D. Zhong, and S. Qu, Modeling the mechanical behaviors of multiple network elastomers, *Mech. Mater.* **161**, 103992 (2021).
 - 18 C. Y. Hui, and R. Long, A constitutive model for the large deformation of a self-healing gel, *Soft Matter* **8**, 8209 (2012).
 - 19 R. W. Ogden, and D. G. Roxburgh, A pseudo-elastic model for the Mullins effect in filled rubber, *Proc. R. Soc. Lond. A* **455**, 2861 (1999).
 - 20 D. Simulia, Abaqus 6.11 analysis user's manual, 2011.
 - 21 L. Boltzmann, Zur theorie der elastischen nachwirkung, *Annalen der Physik* **241**, 430 (1878).
 - 22 B. D. Coleman, and W. Noll, Foundations of linear viscoelasticity, *Rev. Mod. Phys.* **33**, 239 (1961).
 - 23 R. Christensen, Theory of Viscoelasticity: An Introduction (Academic Press, New York, 1982).
 - 24 M. Kaliske, and H. Rothert, Formulation and implementation of three-dimensional viscoelasticity at small and finite strains, *Comput. Mech.* **19**, 228 (1997).
 - 25 J. Shim, and D. Mohr, Rate dependent finite strain constitutive model of polyurea, *Int. J. Plast.* **27**, 868 (2011).
 - 26 J. C. Simo, On a fully three-dimensional finite-strain viscoelastic damage model: Formulation and computational aspects, *Comput. Methods Appl. Mech. Eng.* **60**, 153 (1987).
 - 27 M. S. Green, and A. V. Tobolsky, A new approach to the theory of relaxing polymeric media, *J. Chem. Phys.* **14**, 80 (1946).
 - 28 F. Sidoroff, Un modèle viscoélastique non linéaire avec configuration intermédiaire, **13**, 674 (1974).
 - 29 F. Sidoroff, Incremental constitutive equation for large strain elastoplasticity, *Int. J. Eng. Sci.* **20**, 19 (1982).
 - 30 S. Reese, and S. Govindjee, A theory of finite viscoelasticity and numerical aspects, *Int. J. Solids Struct.* **35**, 3455 (1998).
 - 31 J. Bergstrom, and M. Boyce, Constitutive modeling of the large strain time-dependent behavior of elastomers, *J. Mech. Phys. Solids* **46**, 931 (1998).
 - 32 J. S. Bergström, and M. C. Boyce, Large strain time-dependent behavior of filled elastomers, *Mech. Mater.* **32**, 627 (2000).
 - 33 S. Reese, A micromechanically motivated material model for the thermo-viscoelastic material behaviour of rubber-like polymers, *Int. J. Plast.* **19**, 909 (2003).
 - 34 S. Wang, M. Decker, D. L. Henann, and S. A. Chester, Modeling of dielectric viscoelastomers with application to electromechanical instabilities, *J. Mech. Phys. Solids* **95**, 213 (2016).
 - 35 J. Zhou, L. Jiang, and R. E. Khayat, A micro-macro constitutive model for finite-deformation viscoelasticity of elastomers with nonlinear viscosity, *J. Mech. Phys. Solids* **110**, 137 (2018).
 - 36 S. Özüpek, and E. B. Becker, Constitutive equations for solid propellants, *J. Eng. Mater. Tech.* **119**, 125 (1997).
 - 37 B. Tunç, and Ş. Özüpek, Constitutive modeling of solid propellants for three dimensional nonlinear finite element analysis, *Aerosp. Sci. Tech.* **69**, 290 (2017).
 - 38 B. Tunç, and Ş. Özüpek, Implementation and validation of a three dimensional damaging finite strain viscoelastic model, *Int. J. Solids Struct.* **102-103**, 275 (2016).
 - 39 M. Sasso, G. Chiappini, M. Rossi, L. Cortese, and E. Mancini, Visco-hyper-pseudo-elastic characterization of a fluoro-silicone rubber, *Exp. Mech.* **54**, 315 (2014).
 - 40 T. Lu, J. Wang, R. Yang, and T. J. Wang, A constitutive model for soft materials incorporating viscoelasticity and Mullins effect, *J. Appl. Mech.* **84**, 021010 (2017).
 - 41 T. Lu, Z. Wang, J. Tang, W. Zhang, and T. Wang, A pseudo-elasticity theory to model the strain-softening behavior of tough hydrogels, *J. Mech. Phys. Solids* **137**, 103832 (2020).
 - 42 J. E. Marsden, and T. J. Hughes, Mathematical Foundations of Elasticity (Courier Corporation, 1994).
 - 43 E. Stein, and G. Sagar, Convergence behavior of 3D finite elements for Neo-Hookean material, *Eng. Comput.* **25**, 220 (2008).
 - 44 K. Alkhoury, N. Bosnjak, Y. Wang, H. Lee, S. Nadimpalli, and S. A. Chester, Experiments and modeling of the thermo-mechanically coupled behavior of VHB, *Int. J. Solids Struct.* **242**, 111523 (2022).
 - 45 S. Wang, and S. A. Chester, Experimental characterization and continuum modeling of inelasticity in filled rubber-like materials, *Int. J. Solids Struct.* **136-137**, 125 (2018).

橡胶类材料有限变形黏-超弹性损伤模型：理论和数值实现

杜振江, 杨彦, 王仲彤, 范兴贵, 卢同庆

摘要 橡胶类材料存在着超弹性、时间相关性、率相关性和渐进损伤行为。作者之前的工作提出了一种超弹性损伤模型去描述软材料的应变软化行为(Lu et al., 2020)。模型在单根分子链应变能函数中引入损伤变量 D , 然后将宏观变形和单链分子的伸长率联系在一起。本文改进了之前的模型, 通过引入以Prony级数为基础的非线性黏弹性理论使得模型可以表征时间相关黏性效应。本文进一步将建立的有限变形黏-超弹性损伤本构模型在有限元软件ABAQUS中通过用户自定义子程序UMAT实现。本文利用文献中丙烯酸聚合物在单轴拉伸下的实验数据来校准本构模型的10个参数: 包括4个弹性参数和6个黏性参数。利用ABAQUS模拟丙烯酸聚合物复杂几何形状试样的单轴拉伸和应力松弛, 仿真结果与实验数据一致性较好。